# Agricultural Pesticide Sales Intensity and Female Breast Cancer Mortality Across Mainland French Departments: An Ecological Correlational Study

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## Abstract

**Objective**: To test whether department-level agricultural pesticide sales intensity is associated with age-standardized female breast cancer mortality across mainland France, adjusting for socioeconomic deprivation and population density. **Method**: Ecological (area-level) cross-sectional design. Pesticide sales intensity (kg active substance sold per hectare of utilized agricultural area per year, 2008–2014, log-transformed) was drawn from the Banque Nationale des Ventes de produits phytopharmaceutiques (BNV-D), normalized by Agreste agricultural-census data. The outcome was the age-standardized female breast cancer mortality rate (per 100,000, 2007–2014) from Santé publique France/Francim-InVS-HCL modeled department estimates. Confounders were department-level poverty rate and log population density (INSEE). After excluding overseas departments and four inner-Paris-region departments with near-zero agricultural-area denominators (a data-quality exclusion made without reference to outcome values), the analysis sample was N = 91 mainland departments. The pre-registered model was OLS with HC3-robust standard errors. **Results**: Pesticide sales intensity was positively but non-significantly associated with breast cancer mortality (β = 0.158, 95% CI [-0.126, 0.443], SE = 0.145, t(87) = 1.09, p = .275; partial r = .101, 95% CI [-0.110, 0.303]). The observed effect (partial r = .101) fell below the study's own minimum detectable effect at 80% power (partial r ≈ .288), so the null result is reported as "no effect detected above partial r ≈ .29," not as evidence of no association. The direction was positive in four of five robustness/sensitivity checks; only an exploratory purchase-based exposure specification reached nominal significance (p = .040). Residuals showed significant spatial autocorrelation (Moran's I = .83, p = .001), more consistent with the outcome's own model-smoothed construction than with genuine unmodeled spatial confounding; spatial-lag and spatial-error refits further attenuated the coefficient (both

non-significant). **Conclusion**: This ecological analysis found no statistically detectable association between department-level pesticide sales intensity and female breast cancer mortality in mainland France, in a design underpowered relative to the smallest comparable prior finding. No prior study was found testing this specific pairing, so this null result addresses a genuine gap in the literature rather than replicating an existing one. Findings are associational only and cannot support causal or individual-level claims.

**Keywords**: pesticides, breast cancer, ecological study, France, environmental epidemiology, mortality

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## Introduction

Breast cancer is the most commonly diagnosed cancer among women in France, and established individual-level risk factors (age, reproductive history, family history, hormonal exposure) explain at best roughly 30% of cases, motivating sustained interest in environmental contributors (Davis et al. 1993). Agricultural pesticides have long been hypothesized as one such contributor via the xenoestrogen hypothesis: several organochlorine compounds, polycyclic aromatic hydrocarbons, and triazine herbicides are estrogenically active and induce mammary carcinogenesis in experimental models (Davis et al. 1993), and a broader review concludes that epidemiological and molecular evidence supports excess cancer risk among pesticide applicators and other exposed populations (Alavanja, Ross, and Bonner 2013).

The individual-level (biomarker) literature on pesticide exposure and breast cancer shows a real, but modest and inconsistent, association concentrated in DDT-family compounds. The most current synthesis identified, a 2024 systematic review and meta-analysis of serum and adipose DDT metabolite concentrations, found summary odds ratios ranging from 1.15 to 2.24 depending on the specific metabolite, exposure window, and study design (Ugalde-Resano et al. 2024). A broader review of environmental chemicals and breast cancer reports comparable risk-estimate ranges (2.14–5.0) for DDT, dioxins, PFOSA, and air pollution specifically when exposure occurs during biologically sensitive windows of mammary development — in utero, puberty, or pregnancy — and notes that most epidemiological studies do not capture this exposure timing, a major source of heterogeneity in the literature (Rodgers et al. 2018).

However, this literature is genuinely mixed. Early descriptive and cohort studies found strong positive associations between DDT/DDE/dieldrin and breast cancer that most later case-control studies failed to replicate (Snedeker 2001). A large NHANES-based cross-sectional study found no association between serum organochlorine concentrations and breast cancer prevalence, while finding a significant association for prostate cancer in the same sample (Xu et al. 2009). Most relevant to the present design, the study closest in spirit to an area-level exposure-proxy approach — a GIS-based analysis of residential proximity to agricultural pesticide applications in the 114,835-woman California Teachers Study cohort — found no association between proximity-based exposure and breast cancer incidence across probable/likely carcinogens, mammary carcinogens, or endocrine disruptors, with no dose-response

trend (hazard ratios 1.03–1.15, all confidence intervals crossing 1) (Reynolds et al. 2004). A large prospective cohort of licensed pesticide applicators similarly found no association between glyphosate use and cancer overall (Andreotti et al. 2018). Area- or proximity-based exposure-proxy designs — the design family the present study belongs to — have therefore tended to find weaker or null signals relative to biomarker-based designs, and contextual (area-level) effects on health outcomes are generally modest relative to compositional (individual-level) effects (Pickett and Pearl 2001).

In France specifically, the literature is thin. The AGRICAN cohort of French agricultural workers, using an individual-level crop-exposure matrix (PESTIMAT), has produced pesticide-cancer findings for central nervous system tumors (Piel et al. 2019) and Parkinson's disease (Pouchieu et al. 2018), but no breast-cancer-specific AGRICAN analysis was identified. The best-documented pesticide-cancer story involving a French population is chlordecone contamination in Martinique and Guadeloupe, but the associated outcome is prostate cancer, and the affected departments are overseas territories with a distinct, legacy organochlorine exposure context rather than mainland agricultural sales patterns (Multigner et al. 2010, 2015; Landau-Ossondo et al. 2009). One paper models the spatialization of French pesticide sales data (BNV-D-type data) for environmental-policy monitoring, confirming this data source has been used for department/commune-level spatial modeling in France, but for environmental rather than health outcomes (Martin et al. 2023). No study was found linking mainland-France department-level agricultural pesticide use or sales to breast cancer incidence or mortality — a genuine gap this study addresses, not merely a search-coverage artifact, based on a structured scan of OpenAlex, Semantic Scholar, and Crossref.

Ecological (area-level) designs carry well-documented methodological cautions that motivated specific design choices here. Small-area cancer analyses face a large random component in rates and reduced power to detect small-area effects, with confounding — particularly by socioeconomic factors — a central concern (Elliott and Wartenberg 2004). County-level US evidence confirms that socioeconomic deprivation and rural-urban/urbanization status are independently associated with cancer mortality, with deprivation the stronger driver and steeper gradients in non-metropolitan areas (Singh et al. 2011), supporting the decision to include both deprivation and population density as separate, co-adjusted confounders rather than collapsing them. Breast cancer incidence has also been shown to cluster geographically at the city level elsewhere (Moran's I significant in Saudi Arabia) (Al-Ahmadi and Al-Zahrani 2013), motivating an explicit test for spatial autocorrelation rather than an assumption of independent department-level errors. Finally, area-level associations cannot support individual-level causal claims — the standard ecological-fallacy caution (Blakely and Woodward 2000; Loney and Nagelkerke 2014) — so this study uses associational language throughout and does not interpret its findings causally.

Based on this literature, the following pre-registered, directional hypothesis was tested (verbatim from the approved analysis plan):

**H1 (confirmatory, primary)**: Department-level agricultural pesticide sales intensity (kg active substance / ha utilized agricultural area, 2008–2014 average) is positively associated with age-standardized female breast cancer mortality rate (2007–2014), adjusting for poverty rate and population density.

Three additional questions were pre-specified as exploratory, not subject to the confirmatory alpha: (E1) whether the association holds using purchase-based rather than distributor-sales-based exposure; (E2) whether the association direction replicates in the subsample of departments with directly observed cancer-registry incidence data; and (E3) whether spatial autocorrelation is present in the model residuals and, if so, how a spatial-lag/error model compares to the primary OLS estimate.

# Method

## Data Source and Sample

This is a single-cross-section ecological (area-level) study; the unit of analysis is the mainland French department. Four data sources were combined at department grain. The outcome — age-standardized female breast cancer mortality rate (ICD-10 equivalent "Sein," /100,000, 2007–2014 pooled period) — came from the Santé publique France / Francim-InVS-HCL modeled department estimates (Odissé open-data portal), a model-smoothed rather than raw-registry-count series with full coverage of all French departments. The exposure — pesticide sales intensity — was constructed from BNV-D (Banque Nationale des Ventes de produits phytopharmaceutiques) department-level active-substance sales, summed over 2008–2014 and normalized by Agreste utilized agricultural area (Surface Agricole Utilisée, SAU). Confounders were department-level poverty rate (FiLoSoFi, % below 60% of median income) and population density (inhabitants/km$^2$), both from INSEE via L'Observatoire des Territoires.

Sample construction proceeded as follows (full detail and exact department-level counts in `results/sample_flow.md`). The outcome file provided N = 98 department records including overseas departments (DOM); excluding DOM per the mainland-only inclusion criterion left N = 95. Merging exposure, the SAU normalizer, and both confounders onto this mainland set produced zero unmatched records at every join step (N = 95 throughout). Constructing the pesticide-intensity variable (sales ÷ SAU) revealed that four inner-Paris-region departments — Paris, Hauts-de-Seine, Seine-Saint-Denis, and Val-de-Marne — have utilized agricultural areas so close to zero (1, 5, 526, and 1,039 hectares respectively, versus a next-smallest value of 19,689 hectares for a genuinely rural department) that dividing nonzero recorded BNV-D sales by these near-zero denominators produced implausible intensity values (up to ~21,000 kg/ha/year). This pattern is consistent with the documented BNV-D limitation that sales are recorded at the distributor's department rather than the department of agricultural use, and national wholesaler head offices are disproportionately located in the Paris region. A fixed threshold (SAU < 10,000 ha), set from a large natural gap in the SAU distribution and applied without reference to outcome values, excluded exactly these four departments, yielding a final analysis sample of **N = 91 mainland departments**. No further missingness was present in the complete-case dataset (0/91 departments missing any of the four analysis variables), so no imputation was required.

Because this pesticide-intensity variable could only be meaningfully constructed for genuinely agricultural departments, findings should be understood as applying to mainland France's agricultural departments rather than to fully urbanized areas, where the sales-based exposure measure breaks

down for reasons of data construction rather than substantive irrelevance.

## Measures

**Outcome**: age-standardized (world standard population) female breast cancer mortality rate per 100,000 population, department level, pooled 2007–2014 period (M = 15.37, SD = 1.36, range 13.02–20.77; Table 1).

**Exposure**: pesticide sales intensity, defined as total kg of active substance sold (summed 2008–2014, from BNV-D `VENTE_DPT_SUBSTANCE` files), divided by 7 to annualize, and divided by department SAU (hectares, Agreste RA2020 vintage — the planned RA2010 vintage was not available via any confirmed-working endpoint; this is a logged deviation, see Limitations). The primary operationalization is natural-log-transformed (M = 0.58, SD = 1.01 in log units), reflecting the right-skewed distribution typical of agricultural intensity variables; the untransformed linear version (M = 2.76 kg/ha/year, SD = 2.95) was retained for a robustness check.

**Confounders**: poverty rate (% of population below 60% of median income, FiLoSoFi, 2020 vintage — the planned ~2012 vintage was not available at department grain without an unaudited commune-to-department aggregation; logged deviation) and log population density (INSEE, 2011 vintage; M = 4.45, SD = 0.82 in log inhabitants/km$^2$).

Corsica's two modern department codes (2A, 2B) were aggregated into a single unified code to match the outcome data's coding convention (summed for extensive quantities — sales, SAU; unweighted mean for rate quantities — poverty rate, density), affecting one of the 91 final analysis units.

## Analysis Plan

Analyses were pre-specified in a preregistration-style plan (`analysis/plan.md`) written and approved before any analysis touching outcome data, per the study's governing protocol. The pre-specified primary model was ordinary least squares regression,

$$\text{mortality_rate} = \beta_0 + \beta_1 \log(\text{pesticide_intensity}) + \beta_2 \, \text{poverty_rate} + \beta_3 \log(\text{pop_density}) + \varepsilon$$

with HC3 heteroskedasticity-robust standard errors as the pre-specified baseline (chosen given small-N department-level data with heterogeneous population sizes), a two-sided alpha of .05 on the pesticide coefficient, and no multiple-comparison correction (H1 was the sole confirmatory hypothesis). Effect sizes are reported as the standardized regression coefficient and as a partial correlation with a 95% Fisher-transformed confidence interval, for comparability with the correlational literature.

A full pre-specified diagnostic battery was run regardless of outcome: linearity (partial-residual plot), heteroskedasticity (Breusch-Pagan test, though HC3 SEs were used as baseline regardless of result), normality of residuals (Shapiro-Wilk, Q-Q plot), multicollinearity (variance inflation factors), and influential-point detection (Cook's distance, threshold 4/n), with a pre-specified contingency

(Huber-T robust regression and Spearman correlation) triggered if residuals proved significantly non-normal. Spatial autocorrelation in model residuals was tested via Moran's I using a Queen-contiguity spatial weights matrix built from department boundary polygons, with a pre-specified contingency (spatial-lag and spatial-error maximum-likelihood models) triggered if Moran's I proved significant.

A power analysis was computed programmatically at the achieved sample size (N = 91, residual df = 87) using `statsmodels`' F-test power routine: at 80% power and alpha = .05, this design can detect a minimum effect of partial r ≈ .288 (Cohen's $f^2$ ≈ .090). This is a materially larger effect than the null-to-small hazard ratios (1.03–1.15) found in the closest area-level design analog in the literature (Reynolds et al. 2004), meaning the design is underpowered relative to the smallest comparable prior finding; a non-significant result was therefore pre-specified to be reported as "no effect detected above partial r ≈ .29," not as evidence of no association.

Four robustness checks were pre-specified: (1) an alternative, purchase-based (buyer postal code) exposure measure in place of the primary distributor-sales-based measure; (2) an untransformed (linear, non-log) exposure specification; (3) the FRANCIM registry-department incidence subsample (E2, evaluated for sign/direction consistency only, given its severe power limitation at N ≈ 20); and (4) comparison of the OLS estimate against the spatial-lag/error models triggered by a significant Moran's I.

All analyses were conducted in Python 3.12 using pandas, numpy, scipy, statsmodels, pingouin, libpysal, esda, and spreg (exact pinned versions in `requirements.txt`), with a fixed random seed (42) throughout. A fresh execution of `analysis/run_all.py` reproduces every number reported below.

# Results

## Descriptive Statistics

**Table 1**

*Descriptive Statistics for Analysis Variables (N = 91 Mainland French Departments)*

| Variable | N | Missing | M | SD | Min | Max |
|---|---|---|---|---|---|---|
| Breast cancer mortality rate (age-std., /100,000) | 91 | 0 | 15.37 | 1.36 | 13.02 | 20.77 |
| Pesticide sales intensity, linear (kg a.s./ha/year) | 91 | 0 | 2.76 | 2.95 | 0.04 | 19.81 |

| Variable | N | Missing | M | SD | Min | Max |
|---|---|---|---|---|---|---|
| Pesticide sales intensity, log (ln kg a.s./ha/year) | 91 | 0 | 0.58 | 1.01 | -3.17 | 2.99 |
| Poverty rate (%, 2020 vintage) | 91 | 0 | 14.14 | 2.60 | 8.9 | 20.7 |
| Population density, linear (in-hab./km$^2$, 2011 vintage) | 91 | 0 | 125.87 | 147.59 | 14.93 | 946.94 |
| Population density, log (ln inhab./km$^2$, 2011 vintage) | 91 | 0 | 4.45 | 0.82 | 2.70 | 6.85 |

Bivariate Pearson correlations among primary model variables were modest: mortality with log pesticide intensity, $r$ = .164; mortality with poverty rate, $r$ = .156; mortality with log population density, $r$ = .087; log pesticide intensity correlated with log population density at $r$ = .486, the largest inter-predictor correlation observed, though variance inflation factors in the primary model remained low (all < 1.5), indicating no multicollinearity concern.

## H1: Primary Confirmatory Analysis

H1 was **not supported** at the pre-specified alpha = .05 threshold. The estimated association between pesticide sales intensity and breast cancer mortality was positive but not statistically significant: β = 0.158, 95% CI [-0.126, 0.443], SE = 0.145, t(87) = 1.092, p = .275, standardized β = 0.118, partial r = .101, 95% CI [-0.110, 0.303] (p = .348, classical/non-HC3 inference; this p-value differs slightly from the coefficient's HC3-robust p-value because the two use different variance estimators for the same underlying association). The full model fit was modest: $R^2$ = .044, adjusted $R^2$ = .011, F(3, 87) = 1.056, p = .372. Neither confounder reached significance (poverty rate: b = 0.071, 95% CI [-0.073, 0.215], p = .334; log population density: b = 0.067, 95% CI [-0.427, 0.562], p = .789). Figure 1 displays the bivariate relationship between pesticide sales intensity and breast cancer mortality.

At the achieved sample size, this design had 80% power to detect a minimum effect of partial r ≈ .288

(Cohen's $f^2 \approx .090$); the observed partial r (.101) fell below this threshold. Consistent with the pre-registered interpretation rule, this result is reported as **no effect detected above partial r ≈ .29**, not as evidence of no association, because the design is underpowered relative to the smallest comparable prior finding in the literature (Reynolds et al. 2004).

**Assumption diagnostics.** The Breusch-Pagan test indicated heteroskedasticity (LM = 17.46, p < .001), consistent with the pre-specified use of HC3-robust standard errors as the baseline estimator regardless of this result. Residuals departed significantly from normality (Shapiro-Wilk W = .945, p < .001), triggering the pre-specified heavy-tailed contingency: a Huber-T robust regression produced a similar, still non-significant, coefficient (b = 0.191, p = .194), and the raw bivariate Spearman correlation was ρ = .185 (p = .079). Five departments (Nord, Pyrénées-Orientales, Var, Alpes-Maritimes, Pas-de-Calais) exceeded the Cook's distance influence threshold (4/n = .044); a sensitivity model excluding them produced a larger, nominally significant coefficient (b = 0.270, p = .024, n = 86) — reported here as a sensitivity result alongside, not in place of, the primary estimate, since these departments were not excluded from the pre-registered primary analysis.

**Spatial autocorrelation.** Moran's I on the primary model's residuals (Queen contiguity, N = 91) was significant (I = .829, expected I under the null = -.011, p = .001, 999 permutations), triggering the pre-specified spatial-lag/error contingency. However, I = .829 is far above the range typically observed for genuine areal disease-risk clustering in the epidemiological literature (roughly .2–.5); combined with a spatial-error lambda of .969 (near-perfect predictability of neighboring residuals) alongside a very low spatial-error pseudo-$R^2$ (.021), this pattern is more consistent with an artifact of the outcome's own model-smoothed construction — the InVS/Francim mortality estimates are spatiotemporally smoothed at the estimation stage, which would mechanically inflate residual spatial autocorrelation independent of any true geographic clustering in breast-cancer risk — than with a genuine omitted spatial confounder. Accordingly, the spatial-lag (b = 0.112, p = .393) and spatial-error (b = 0.080, p = .146) coefficients are reported as exploratory and model-dependent rather than as a validated spatial correction; both are attenuated relative to, and consistent in direction with, the primary OLS estimate, and both remain non-significant, so this diagnostic does not overturn the primary null finding.

## Robustness Checks

**Table 2**

*Robustness of the Pesticide Coefficient Across Pre-Specified Alternative Specifications*

| Check | Coefficient | 95% CI | p | N |
|---|---|---|---|---|
| H1 primary (sales-based, log exposure) | 0.158 | [-0.126, 0.443] | .275 | 91 |

| Check | Coefficient | 95% CI | p | N |
|---|---|---|---|---|
| 1. Purchase-based exposure | 0.276 | [0.012, 0.541] | .040 | 91 |
| 2. Linear (non-log) exposure | -0.045 | [-0.158, 0.067] | .432 | 91 |
| 4a. Spatial lag (vs. OLS) | 0.112 | [-0.145, 0.369] | .393 | 91 |
| 4b. Spatial error (vs. OLS) | 0.080 | [-0.028, 0.189] | .146 | 91 |

The pesticide coefficient's direction was positive — consistent with the primary estimate — in four of five specifications; it reversed sign only under the untransformed linear-exposure specification (check 2), which was itself non-significant in either direction. Only one specification, purchase-based exposure (check 1), reached nominal significance (p = .040); because this was not the pre-registered primary specification, it is reported at face value as an exploratory finding rather than treated as confirmatory (see Figure 2 for a visual forest-plot summary).

## Exploratory Analyses

*The following analyses were labeled exploratory in the pre-registered plan and are not subject to the confirmatory alpha; p-values are reported nominally, for reference only.*

**E1 (alternative exposure measure).** Substituting purchase-based (buyer postal code, 2013–2023) exposure for the primary distributor-sales-based measure produced a larger, nominally significant coefficient (b = 0.276, 95% CI [0.012, 0.541], p = .040, N = 91), matching the primary estimate's direction.

**E2 (registry-observed incidence subsample).** In the N = 19 FRANCIM registry-covered departments, using directly observed breast cancer incidence (2018–2022) rather than modeled mortality, the coefficient was positive and matched the primary direction (b = 0.441, 95% CI [-1.892, 2.774], p = .711) but was not independently powered (minimum detectable partial r ≈ .58 at this sample size) and is interpreted for sign/direction consistency only.

**E3 (spatial autocorrelation).** As detailed above, Moran's I on H1 residuals was significant (I = .829, p = .001), and the triggered spatial-lag/error models produced attenuated, non-significant pesticide coefficients, most plausibly reflecting the outcome's own smoothed construction rather than genuine spatial confounding.

## Discussion

This ecological, department-level analysis found no statistically detectable association between agricultural pesticide sales intensity and female breast cancer mortality across 91 mainland French departments (partial r = .10, 95% CI [-0.11, 0.30], p = .35), a design that was, by its own pre-specified power analysis, underpowered to detect an effect as small as the null-to-modest hazard ratios (1.03–1.15) found in the most methodologically comparable prior study, a proximity-based analysis in the California Teachers Study cohort (Reynolds et al. 2004). This result should therefore be read as "no effect detected above a moderate threshold," not as evidence that no association exists. No prior study — ecological or individual-level — was identified linking mainland-France department-level pesticide use to breast cancer outcomes, so this null finding fills a genuine gap in the literature rather than failing to replicate an established one.

The finding is broadly consistent with the pattern observed across the area-level exposure-proxy literature more generally: while individual-level biomarker studies of DDT-family compounds show modest positive associations (summary odds ratios 1.15–2.24 depending on compound and design) (Ugalde-Resano et al. 2024), proximity- or area-based exposure-proxy designs — several steps further removed from individual internal dose — have tended to find weaker or null signals (Reynolds et al. 2004; Andreotti et al. 2018), consistent with evidence that contextual (area-level) effects on health outcomes are generally modest relative to compositional (individual-level) effects (Pickett and Pearl 2001). The present department-level design, several further steps removed from individual exposure than even a proximity-based measure, was expected on these grounds to detect at most an attenuated signal, and the observed result is consistent with that expectation.

Robustness checks broadly, though not uniformly, supported the primary finding's direction: the pesticide coefficient was positive in four of five specifications, and only one alternative specification (a purchase-based exposure measure, more closely tied to point of agricultural use than distributor-based sales) reached nominal significance. Because this was an exploratory, non-primary specification, it is reported as a signal worth future investigation rather than as a confirmed finding, and its emergence alongside four other broadly consistent-direction (if non-significant) estimates is more suggestive of a real, small, currently underpowered signal than of a spurious primary result — though this interpretation remains speculative given the study's power limitations.

### Limitations

Several limitations qualify these findings. First, and most fundamentally, this is an **ecological, area-level design**: department-level correlations between aggregate pesticide sales and aggregate mortality cannot support individual-level causal or even individual-level correlational claims, and are vulnerable to the ecological fallacy — the true individual-level relationship between pesticide exposure and breast cancer risk could be stronger, weaker, or differently confounded than the department-level pattern observed here (Blakely and Woodward 2000; Loney and Nagelkerke 2014). Second, the exposure measure captures **recent agricultural sales intensity (2008–2014)**, not the biologically relevant exposure window; several of the strongest individual-level findings in the literature depend on

exposure occurring decades before diagnosis, often during in-utero, pubertal, or pregnancy windows (Rodgers et al. 2018), which this department-level sales proxy cannot capture. Third, BNV-D sales are recorded at the **distributor's department**, not necessarily the department of agricultural use — a documented limitation that likely introduces exposure misclassification and, in four inner-Paris-region departments, produced denominator artifacts severe enough to require exclusion from the analysis sample (see Method). Fourth, two confounder/normalizer vintages deviated from the pre-registered plan: the agricultural-area normalizer used the 2020 rather than the planned 2010 agricultural census, and the poverty-rate confounder used a 2020 rather than the planned circa-2012 vintage, in both cases because the originally planned vintage was not available via any confirmed-working public endpoint after reasonable effort — an 8-year gap from the 2007–2014 mortality period for the poverty measure specifically, which somewhat limits, though does not eliminate, its value as a confounder for that period. Fifth, the significant spatial autocorrelation detected in model residuals (Moran's I = .83) is most plausibly attributable to the outcome variable's own model-smoothed construction rather than to a genuine unmodeled spatial confounder, meaning the spatial-lag/error robustness check should not be over-interpreted as a validated correction, though it is also unlikely to be masking a substantively different conclusion given its consistency in direction with the primary estimate. Sixth, the study is **underpowered relative to the smallest comparable prior effect size** in the literature, so the null result constrains only effects at or above a moderate threshold (partial $r \approx .29$) and cannot rule out smaller true associations. Finally, only two confounders (deprivation, urbanization) were included; other plausible confounders identified in the literature — mammography screening participation, reproductive and hormonal factors, obesity, alcohol consumption — were not available at department grain within this study's scope and could not be adjusted for.

## Future Directions

A directly powered replication using the purchase-based (rather than distributor-sales-based) exposure measure, which showed the only nominally significant robustness result here, would be a natural next step, ideally paired with a longer or more recent, non-smoothed mortality or incidence series if one becomes available at department grain. Incorporating mammography screening participation as an additional confounder, and exploring finer geographic resolution (e.g., commune- or canton-level, where feasible) to reduce the area-level averaging inherent to department-grain analysis, would also strengthen future work in this area. Given the observed spatial autocorrelation, a raw (non-smoothed) outcome series would allow a more interpretable spatial-lag/error analysis than was possible here.

## Conclusion

This study found no statistically significant association between department-level agricultural pesticide sales intensity and female breast cancer mortality in mainland France (N = 91 departments), in an analysis that was, by design, underpowered relative to the smallest effect sizes reported in the most comparable prior literature. The result is best read as inconclusive rather than as evidence of no association, and represents, to our knowledge, the first test of this specific pairing in the published or gray literature. Findings are strictly associational and should not be interpreted as evidence of individual-

level causation.

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## References


Al-Ahmadi, Khalid, and Ali Al-Zahrani. 2013. "Spatial Autocorrelation of Cancer Incidence in Saudi Arabia." *International Journal of Environmental Research and Public Health*. https://doi.org/10.3390/ijerph10127207.

Alavanja, Michael C., Matthew K. Ross, and Matthew R. Bonner. 2013. "Increased Cancer Burden Among Pesticide Applicators and Others Due to Pesticide Exposure." *CA: A Cancer Journal for Clinicians*. https://doi.org/10.3322/caac.21170.

Andreotti, Gabriella, Stella Koutros, Jonathan N. Hofmann, et al. 2018. "Glyphosate Use and Cancer Incidence in the Agricultural Health Study." *Journal of the National Cancer Institute*. https://doi.org/10.1093/jnci/djx233.

Blakely, Tony A., and Alistair J. Woodward. 2000. "Ecological Effects in Multi-Level Studies." *Journal of Epidemiology and Community Health*. https://doi.org/10.1136/jech.54.5.367.

Davis, Devra Lee, H. Leon Bradlow, Mary Wolff, et al. 1993. "Medical Hypothesis: Xenoestrogens as Preventable Causes of Breast Cancer." *Environmental Health Perspectives*. https://doi.org/10.1289/ehp.93101372.

Elliott, Paul, and Daniel Wartenberg. 2004. "Spatial Epidemiology: Current Approaches and Future Challenges." *Environmental Health Perspectives*. https://doi.org/10.1289/ehp.6735.

Landau-Ossondo, Muriel, Nadege Rabia, Jacqueline Jos-Pelage, et al. 2009. "Why Pesticides Could Be a Common Cause of Prostate and Breast Cancers in the French Caribbean Island, Martinique. An Overview on Key Mechanisms of Pesticide-Induced Cancer." *Biomedicine & Pharmacotherapy*. https://doi.org/10.1016/j.biopha.2009.04.043.

Loney, Tom, and Nico J. Nagelkerke. 2014. "The Individualistic Fallacy, Ecological Studies and Instrumental Variables: A Causal Interpretation." *Emerging Themes in Epidemiology*. https://doi.org/10.1186/1742-7622-11-18.

Martin, Ramalanjaona, Truche, et al. 2023. "Modelling the Spatialisation of Pesticide Sales to Monitor Environmental Policies in France." *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2023.136880.

Multigner, Luc, Philippe Kadhel, Florence Rouget, et al. 2015. "Chlordecone Exposure and Adverse Effects in French West Indies Populations." *Environmental Science and Pollution Research*. https://doi.org/10.1007/s11356-015-4621-5.

Multigner, Luc, Jean Rodrigue Ndong, Andre Giusti, et al. 2010. "Chlordecone Exposure and Risk of Prostate Cancer." *Journal of Clinical Oncology*. https://doi.org/10.1200/jco.2009.27.2153.

Pickett, Kate E., and Michelle Pearl. 2001. "Multilevel Analyses of Neighbourhood Socioeconomic Context and Health Outcomes: A Critical Review." *Journal of Epidemiology and Community Health*. https://doi.org/10.1136/jech.55.2.111.

Piel, Clementine, Camille Pouchieu, Camille Carles, et al. 2019. "Agricultural Exposures to Carbamate Herbicides and Fungicides and Central Nervous System Tumour Incidence in the Cohort AGRICAN."

*Environment International*. https://doi.org/10.1016/j.envint.2019.05.070.

Pouchieu, Camille, Clementine Piel, Camille Carles, et al. 2018. "Pesticide Use in Agriculture and Parkinson's Disease in the AGRICAN Cohort Study." *International Journal of Epidemiology*. https://doi.org/10.1093/ije/dyx225.

Reynolds, Peggy, Susan Hurley, Debbie E. Goldberg, et al. 2004. "Residential Proximity to Agricultural Pesticide Use and Incidence of Breast Cancer in the California Teachers Study Cohort." *Environmental Research*. https://doi.org/10.1016/j.envres.2004.03.001.

Rodgers, Kathryn M., Julia O. Udesky, Ruthann A. Rudel, et al. 2018. "Environmental Chemicals and Breast Cancer: An Updated Review of Epidemiological Literature Informed by Biological Mechanisms." *Environmental Research*. https://doi.org/10.1016/j.envres.2017.08.045.

Singh, Gopal K., Sarah D. Williams, Mohammad Siahpush, et al. 2011. "Socioeconomic, Rural-Urban, and Racial Inequalities in US Cancer Mortality: Part I–All Cancers and Lung Cancer and Part II–Colorectal, Prostate, Breast, and Cervical Cancers." *Journal of Cancer Epidemiology*. https://doi.org/10.1155/2011/107497.

Snedeker, Suzanne M. 2001. "Pesticides and Breast Cancer Risk: A Review of DDT, DDE, and Dieldrin." *Environmental Health Perspectives*. https://doi.org/10.1289/ehp.01109s135.

Ugalde-Resano, Regina, Brenda Gamboa-Loira, Angel Merida-Ortega, et al. 2024. "Biological Concentrations of DDT Metabolites and Breast Cancer Risk: An Updated Systematic Review and Meta-Analysis." *Reviews on Environmental Health*. https://doi.org/10.1515/reveh-2024-0021.

Xu, Xiaohui, Amy B. Dailey, Evelyn O. Talbott, et al. 2009. "Associations of Serum Concentrations of Organochlorine Pesticides with Breast Cancer and Prostate Cancer in u.s. Adults." *Environmental Health Perspectives* 118 (1): 60–66. https://doi.org/10.1289/ehp.0900919.